

VAPO

Value-Model-Based RL for Advanced Reasoning

60.4

AIME 2024 (avg@32)

Base Model: Qwen2.5-32B | No SFT Data

ByteDance Seed



arXiv:2504.05118

Why Value Models?

Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-Free (GRPO/DAPO)

How it works:

- Compute advantage based on final trajectory reward
- Assign advantage uniformly across all tokens
- Trajectory-level credit assignment

Limitations:

- ✗ Cannot trace step-level impact on returns
- ✗ Uniform assignment misses error sources
- ✗ Subtle errors cause catastrophic failures
- ✗ Higher variance in credit estimation

Value-Model-Based (VAPO)

How it works:

- Token-level credit assignment
- Trace each action's impact on returns
- Lower variance per token vs Monte Carlo

Advantages:

- ✓ Precise error localization in multi-step
- ✓ Lower variance estimates
- ✓ Better generalization across samples
- ✓ Improved sample efficiency

Key Insight: Value models enable token-level precision — critical when subtle step-level errors cause catastrophic failures in complex reasoning.

Three Challenges

Plaguing Value-Model-Based RL in Long-CoT Tasks

1

Value Model Bias

Long trajectories +
bootstrapped learning
cause accumulated bias

Bootstrapping error amplifies
across long reasoning chains

→ Value estimates drift from
true returns over time

2

Heterogeneous Lengths

Different responses have
different bias-variance
tradeoffs

Short response: Low λ = low variance
Long response: High λ = low bias

One λ doesn't fit all!

→ GAE λ must adapt to
response length dynamically

3

Reward Sparsity

Only the final `<eos>` token
receives reward signal

Chain: $\text{step}_1 \rightarrow \text{step}_2 \rightarrow \dots \rightarrow \text{step}_n \rightarrow \text{<eos>}$

↑ ↑ ↑

x x ✓

Reward only at final token

→ 99% of tokens get zero
credit despite critical decisions



VAPO Framework

Integrated Solutions for Value-Model-Based Reasoning RL



Value-Pretraining

Pre-trains value model before RL to reduce initial bias

Addresses Challenge 1: Value Model Bias

Decoupled-GAE

Separates GAE computation for robust value estimation

Addresses Challenge 1: Reduces Bootstrap Error

Length-Adaptive GAE

Novel: Dynamic λ adjustment based on response length

Addresses Challenge 2: Heterogeneous Lengths

Clip-Higher

Asymmetric clipping from DAPO for exploration

Addresses Challenge 3

Token-level Loss

Normalizes loss at token level (not sequence level)

From DAPO

Positive Example LM Loss

Self-imitation learning on correct trajectories

Group-Sampling

Samples multiple responses per prompt

From GRPO

Length-Adaptive GAE

Dynamic λ Adjustment for Variable Response Lengths

NOVEL

Core Insight: GAE λ parameter is dynamically adjusted based on response length

Short responses $\rightarrow$ lower λ (lower variance) | Long responses $\rightarrow$ higher λ (lower bias)

Short Responses

Characteristics:

- Few reasoning steps
 - Lower bootstrap error accumulation
- More stable value estimates

λ Setting:

$\lambda = 0.2$

$\leftarrow$ Lower

Result: Lower Variance

Long Responses

Characteristics:

- Many reasoning steps
 - Bootstrap error accumulates
- Value estimates drift more

λ Setting:

$\lambda = 0.95$

$\leftarrow$ Higher

Result: Lower Bias

Optimization Goal

Balance bias-variance tradeoff dynamically across heterogeneous sequence lengths

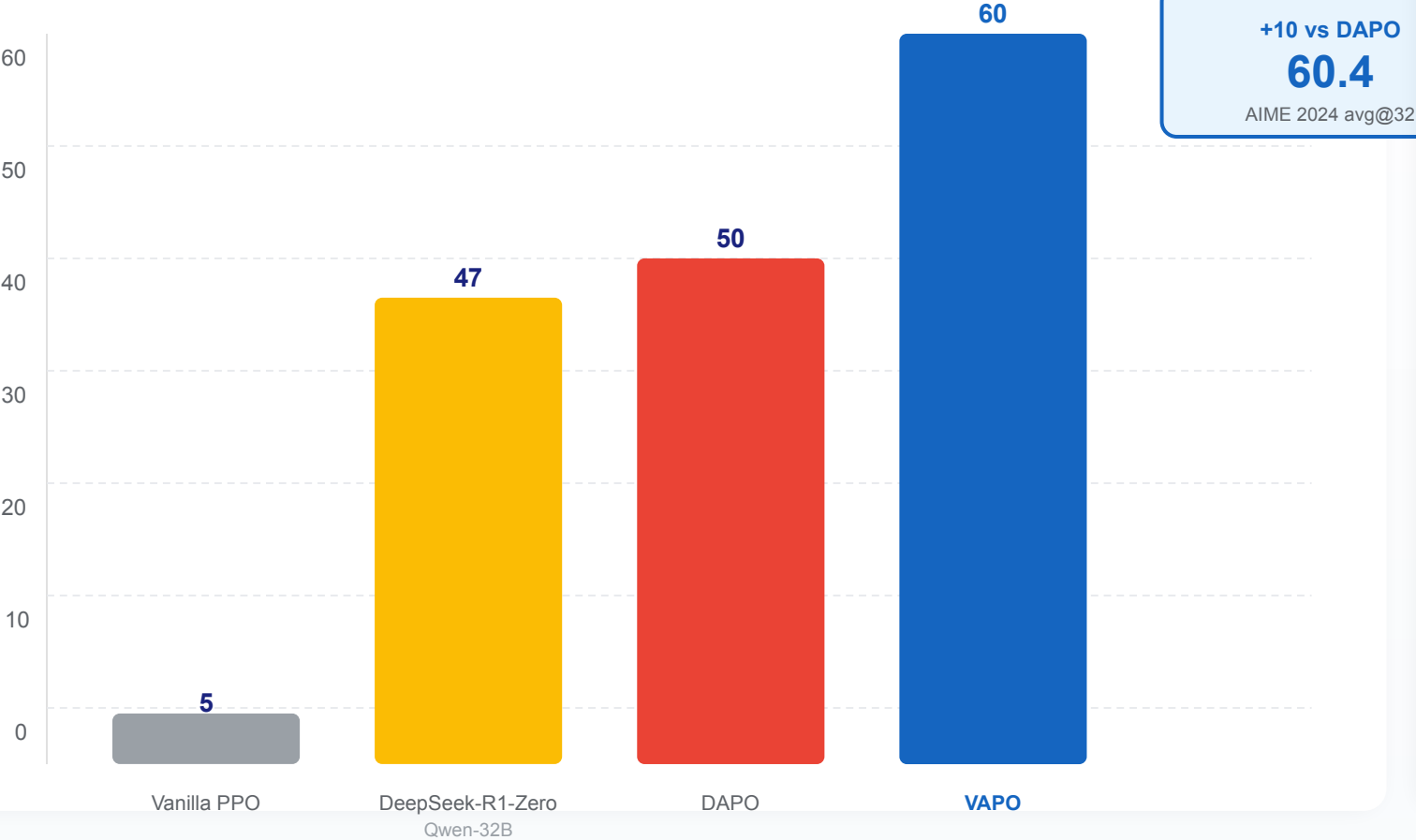
Contributes +15 points to final AIME score



VAPO Scores 60 on AIME 2024

10+ Points Above DAPO and DeepSeek-R1-Zero

AIME 2024 Performance Comparison (avg@32)



Performance Summary

Base Model	Same for all
Qwen2.5-32B	
SFT Data	Pure RL
None	
Training Steps to SOTA	DAPO: ~5,500
~5,000 steps	
Training Stability	Multiple runs
No crashes	
AIME 2024 avg@32	
60.4	
+10.4 vs DAPO (50) +13.4 vs DeepSeek-R1-Zero (47)	

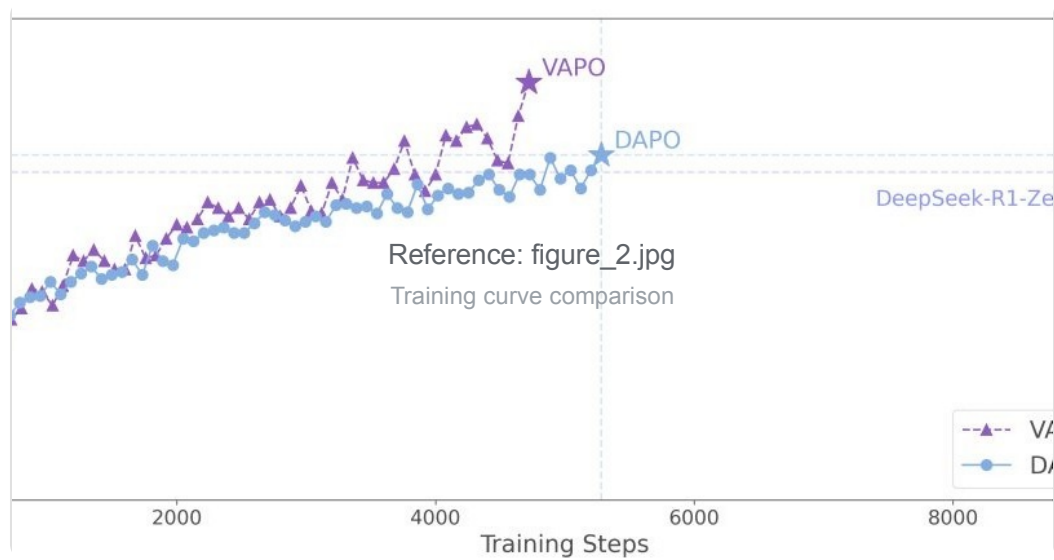
Training curve reference: figure_2.jpg — VAPO consistently outperforms DAPO throughout training



Training Efficiency

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO

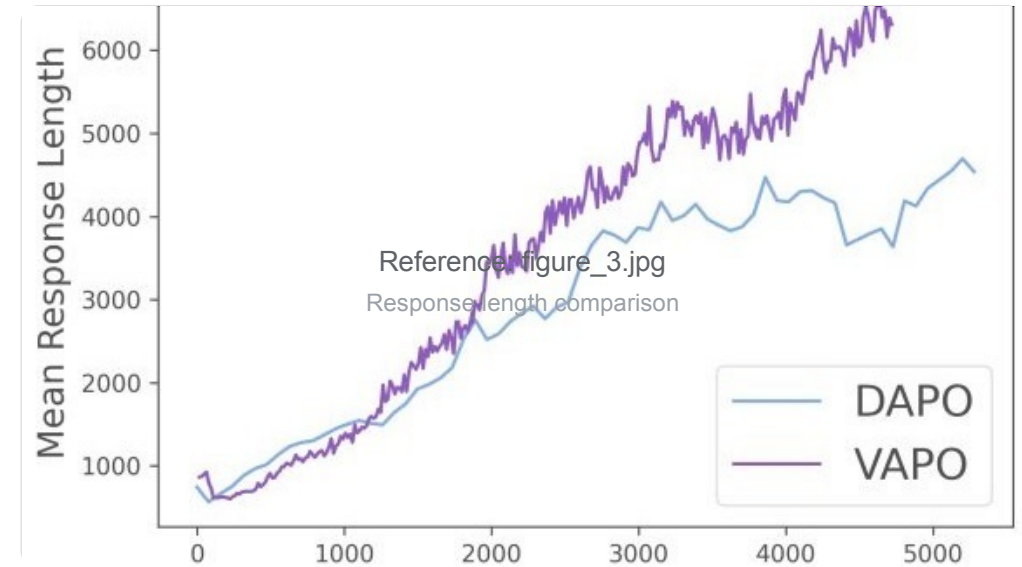
AIME 2024 Accuracy Over Training



VAPO reaches 60%
around step 4,500

DAPO plateaus at 50%
at step 5,500

Mean Response Length Over Training



VAPO: ~6,000 tokens
longer reasoning chains

DAPO: ~4,500 tokens
shorter responses

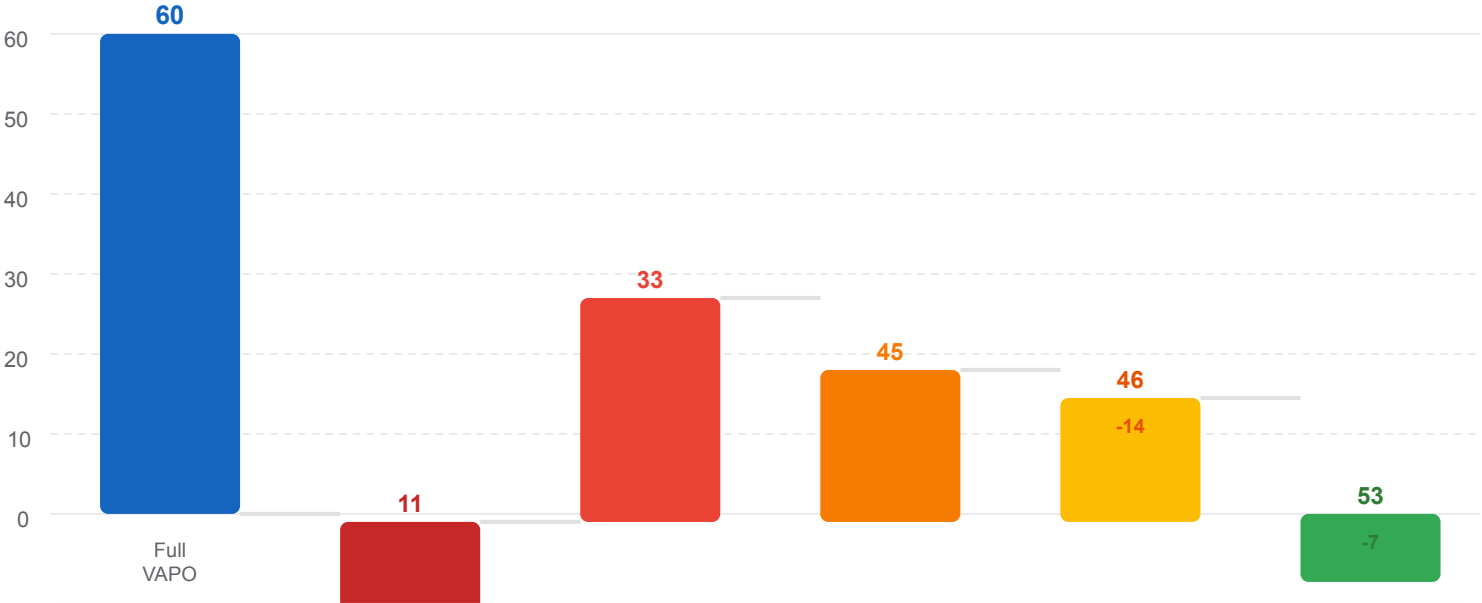
Key Finding: VAPO achieves better accuracy with fewer steps AND generates longer reasoning chains

Superior training efficiency + better reasoning depth = state-of-the-art performance

Ablation Study

Every Component Matters — Value-Pretraining Alone Adds 49 Points

Component Contribution Waterfall



Key Insight: Value-Pretraining is the single most impactful component

Without it, value-model-based RL barely works (drops to 11)

Other components also matter: Decoupled-GAE (-27), Length-adaptive GAE (-15)

Ablation Results

Component	Score	Drop
None (Full VAPO)	60	—
Value-Pretraining	11	-49
Decoupled-GAE	33	-27
Length-adaptive GAE	45	-15
Clip-Higher	46	-14
Token-level Loss	53	-7
Positive Example LM Loss	54	-6
Group-Sampling	55	-5

Key Takeaways

Value-Model-Based RL Is Back — With the Right Engineering

1 Value Pretraining is Critical

+49 points

Contributes the most among all components.

Without it, value-model-based RL barely works.

2 Length-Adaptive GAE

+15 points

Novel technique for heterogeneous sequence lengths.

Dynamic λ adjustment optimizes bias-variance tradeoff.

3 Stable Training

Zero Crashes

No training crashes across multiple independent runs.

Reliable convergence in production environments.

4 Future Implications

Value > Free

Value-model-based RL can surpass value-free methods.

Systematic approach to challenges enables SOTA reasoning.

VAPO Achieves 60.4 on AIME 2024

Same Qwen2.5-32B base model | No SFT Data | Pure RL

ByteDance Seed | arXiv:2504.05118

